

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
```

```
int main() {
```

```
    int cost1, qty1, cost2, qty2, cost3, qty3;
```

```
    int budget;
```

```
    double totalBeforeTax, totalAfterTax;
```

```
    scanf("%d %d", &cost1, &qty1);
```

```
    scanf("%d %d", &cost2, &qty2);
```

```
    scanf("%d %d", &cost3, &qty3);
```

```
    scanf("%d", &budget);
```

```
    totalBeforeTax = (cost1 * qty1) + (cost2 * qty2) + (cost3 * qty3);
```

```
    totalAfterTax = totalBeforeTax + (totalBeforeTax * 0.10);
```

```
    printf("Total (Before Tax): %.2f\n", totalBeforeTax);
```

```
    printf("Total (After Tax): %.2f\n", totalAfterTax);
```

```
    if (totalAfterTax <= budget) {
```

```
        printf("Within Budget");
```

```
    } else {
```

```
        printf("Over Budget");
```

```
    }
```

```
    return 0;
```

```
}
```

1.#Smart Shopper 1

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <math.h>
```

```
#include <stdlib.h>
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b;
```

```
    if (scanf("%d %d", &a, &b) != 2) {
```

```
        return 1;
```

```
    }
```

```
    printf("Sum: %d\n", a + b);
```

```
    printf("Difference: %d\n", a - b);
```

```
    printf("Product: %lld\n", (long long)a * b);
```

```
    printf("Quotient: %d\n", a / b);
```

```
    printf("Remainder: %d\n", a % b);
```

```
    return 0;
```

```
}
```

2.#Basic Arithmetic Operations 8

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <math.h>
```

```
#include <stdlib.h>
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    if (scanf("%d", &n) != 1) {
```

```
        return 1;
```

```
    }
```

```
    if (n == 1) {
```

```
        printf("0\n");
```

```
    }
```

```
    else if (n % 2 == 0) {
```

```
        printf("%d\n", n / 2);
```

```
    }
```

```
    else {
```

```
        printf("%d\n", n);
```

```
    }
```

```
    return 0;
```

```
}
```

3.#MINIMUM CUTS TO DIVIDE A CIRCLE 2

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
int main() {
```

```

int n;
if (scanf("%d", &n) != 1) {
    return 1;
}
if (n % 2 == 0) {
    printf("true\n");
} else {
    printf("false\n");
}

return 0;
}

```

4.#DIVISOR GAME 2

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

```

```

int main() {
    int a, b, c;

    scanf("%d", &a);
    scanf("%d", &b);
    scanf("%d", &c);

    if ((a == b && a >= c) || (a == c && a >= b) || (b == c && b >= a)) {
        printf("There is a tie for the greatest number");
    }
}

```

```

    }
    else if (a > b && a > c) {
        printf("%d is the greatest", a);
    }
    else if (b > a && b > c) {
        printf("%d is the greatest", b);
    }
    else {
        printf("%d is the greatest", c);
    }

    return 0;
}

```

5.#Greatest of Three Numbers 11

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

int main() {
    char ch;
    if (scanf("%c", &ch) != 1) {
        return 1;
    }
    if (ch >= 'A' && ch <= 'Z') {
        printf("Uppercase Alphabet\n");
    }
}

```

```

else if (ch >= 'a' && ch <= 'z') {
    printf("Lowercase Alphabet\n");
}
else if (ch >= '0' && ch <= '9') {
    printf("Digit\n");
}
else {
    printf("Special Character\n");
}

return 0;
}

```

6.#Character Classifier 1

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

int main() {

    int a, b, c;
    scanf("%d %d %d", &a, &b, &c);

    if (a + b <= c || b + c <= a || c + a <= b) {

        printf("Not a valid triangle");

    }
}

```

```
else {  
  
    if (a == b && b == c) {  
  
        printf("Equilateral Triangle");  
  
    }  
  
    else if (a == b || b == c || a == c) {  
  
        printf("Isosceles Triangle");  
  
    }  
  
    else {  
  
        printf("Scalene Triangle");  
  
    }  
  
}  
  
return 0;  
  
}
```

7.#Triangle Type Detector

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

int main() {

    int year;

    scanf("%d", &year);
    if ((year % 400 == 0) || ((year % 4 == 0) && (year % 100 != 0))) {
        printf("Leap Year");
    } else {
        printf("Not a Leap Year");
    }

    return 0;
}
```

8.#Leap Year Checker 28

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

int main() {
    int N, i;
```

```
scanf("%d", &N);
```

```
for (i = 1; i <= N; i++) {
```

```
    if (N % i == 0) {
```

```
        printf("%d ", i);
```

```
    }
```

```
}
```

```
return 0;
```

```
}
```

9.#Print All Factors

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <math.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int N, i, isPrime = 1;
```

```
    scanf("%d", &N);
```

```
    if (N <= 1) {
```

```
        printf("Not Prime");
```

```
        return 0;
```

```
    }
```

```
    for (i = 2; i * i <= N; i++) {
```

```
        if (N % i == 0) {
```

```
        isPrime = 0;
        break;
    }
}

if (isPrime) {
    printf("Prime");
} else {
    printf("Not Prime");
}

return 0;
}
`10.#Is It a Prime Number? 1
```

2 pages

```
#include <stdio.h>
#include <string.h>
```

```

#include <math.h>
#include <stdlib.h>
#include <stdio.h>

int main() {
    int base, exponent;
    int result = 1, i;

    scanf("%d %d", &base, &exponent);

    for (i = 1; i <= exponent; i++) {
        result = result * base;
    }

    printf("%d", result);

    return 0;
}

```

1.#Manual Power Calculator

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

long long fibonacci(int n) {
    if (n <= 1) {
        return n;
    }
}

```

```

    }

    long long prev = 0; // F(0)
    long long curr = 1; // F(1)
    long long next;

    for (int i = 2; i <= n; i++) {
        next = prev + curr;
        prev = curr;
        curr = next;
    }

    return curr;
}

int main() {
    int n;

    printf("Enter the value of n: ");
    if (scanf("%d", &n) != 1 || n < 0) {
        printf("Please enter a non-negative integer.\n");
        return 1;
    }

    printf("F(%d) = %lld\n", n, fibonacci(n));

    return 0;
}

```

2.#Fibonacci Number 43

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

int tribonacci(int n) {
    if (n == 0) return 0;
    if (n == 1 || n == 2) return 1;
    int t0 = 0;
    int t1 = 1;
    int t2 = 1;
    int tn = 0;

    for (int i = 3; i <= n; i++) {
        tn = t0 + t1 + t2;
        t0 = t1;
        t1 = t2;
        t2 = tn;
    }

    return t2;
}

int main() {
    int n;

    printf("Enter the value of n: ");
```

```

if (scanf("%d", &n) != 1 || n < 0) {
    printf("Please enter a non-negative integer.\n");
    return 1;
}
printf("T(%d) = %d\n", n, tribonacci(n));

return 0;
}

```

3.#Tribonacci Number 3

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>
int findGCD(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}
int countCommonFactors(int a, int b) {
    int gcd = findGCD(a, b);
    int count = 0;
    for (int i = 1; i <= gcd; i++) {

```

```

        if (gcd % i == 0) {
            count++;
        }
    }

    return count;
}

int main() {
    int a, b;

    printf("Enter two positive integers (a b): ");
    if (scanf("%d %d", &a, &b) != 2 || a <= 0 || b <= 0) {
        printf("Invalid input. Please enter positive integers.\n");
        return 1;
    }

    int result = countCommonFactors(a, b);
    printf("Number of common factors: %d\n", result);

    return 0;
}

```

4.#NO OF COMMON FACTORS 1

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

```

```
#include <stdbool.h>

bool isThreeDivisors(int n) {
    int count = 0;
    for (int i = 1; i <= n; i++) {
        if (n % i == 0) {
            count++;
        }
        if (count > 3) {
            return false;
        }
    }
    return count == 3;
}
```

```
int main() {
    int n;

    printf("Enter an integer: ");
    if (scanf("%d", &n) != 1) {
        printf("Invalid input.\n");
        return 1;
    }

    if (isThreeDivisors(n)) {
        printf("true\n");
    } else {
        printf("false\n");
    }
}
```

```
    return 0;
}

5.#THREE DIVISORS 3
```

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

int countDigits(int n) {
    if (n == 0) return 1;

    int count = 0;
    while (n > 0) {
        n = n / 10;
        count++;
    }
    return count;
}

int main() {
    int n;

    printf("Enter a natural number: ");
    if (scanf("%d", &n) != 1 || n < 0) {
        printf("Please enter a non-negative integer.\n");
        return 1;
    }
}
```

```
int result = countDigits(n);  
printf("Number of digits: %d\n", result);  
  
return 0;  
}  
6.#COUNT ME 5
```

```
#include <stdio.h>  
#include <stdlib.h>
```

```
int main() {  
    int N;  
    scanf("%d", &N);  
  
    N = abs(N); // take absolute value  
  
    int max = 0, min = 9;  
  
    // special case when N is 0  
    if (N == 0) {  
        printf("0 0");  
        return 0;  
    }  
  
    while (N > 0) {  
        int digit = N % 10;  
  
        if (digit > max)  
            max = digit;
```

```

        if (digit < min)
            min = digit;

        N /= 10;
    }

    printf("%d %d", max, min);
    return 0;
}

7.#Maximum and Minimum Digit

```

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, reverse = 0, digit;
    scanf("%d", &n);

    int sign = (n < 0) ? -1 : 1;
    n = abs(n);

    while (n != 0) {
        digit = n % 10;
        reverse = reverse * 10 + digit;
        n = n / 10;
    }

    printf("%d\n", reverse * sign);
}

```

```
    return 0;
}

8.#Reverse the number 28
```

```
#include <stdio.h>
```

```
int isPalindrome(int n)
{
    int original = n;
    int reverse = 0, digit;

    if (n < 0)
        return 0;

    while (n != 0) {
        digit = n % 10;
        reverse = reverse * 10 + digit;
        n = n / 10;
    }

    if (original == reverse)
        return 1;
    else
        return 0;
}
```

```
int main()
{
    int n;
```

```

scanf("%d", &n);

if (isPalindrome(n))
    printf("Palindrome\n");
else
    printf("Not Palindrome\n");

return 0;
}

```

9.#Palindrome Number 119

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>
#include <math.h>
#include <stdbool.h>

bool isArmstrong(int N) {
    if (N < 0) return false;
    if (N == 0) return true;

    int temp = N;
    int digits = 0;
    long long sum = 0;
    int countTemp = N;
    while (countTemp > 0) {
        countTemp /= 10;
    }
}

```

```

        digits++;
    }
    temp = N;
    while (temp > 0) {
        int remainder = temp % 10
        sum += (long long)round(pow(remainder, digits));

        temp /= 10;
    }
    return (sum == (long long)N);
}

int main() {
    int N;
    if (scanf("%d", &N) != 1) {
        return 0;
    }

    if (isArmstrong(N)) {
        printf("true\n");
    } else {
        printf("false\n");
    }

    return 0;
}

```

10.#Is it an Armstrong Number? 2

#page 3

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int i,j,a,b;
```

```
    scanf("%d%d",&a,&b);
```

```
    for(i=1;i<=a;i++)
```

```
    {
```

```
        for(j=1;j<=b;j++)
```

```
        {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

4.#Rectangle Pattern 6

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```

scanf("%d",&n);

for(int i=1;i<=n;i++)
{
    for(int j=1;j<=i;j++)
    {
        printf("*");
    }
    printf("\n");
}
return 0;
}

```

5.#Staircase of Stars

```

#include<stdio.h>

int main()
{
    int n;
    scanf("%d",&n);

    for(int i=n;i>=1;i--)
    {
        for(int j=1;j<=i;j++)
        {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}

```

```
}
```

6.#Descending Star Steps

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <math.h>
```

```
#include <stdlib.h>
```

```
#include <stdio.h>
```

```
void printBinaryPattern(int n) {
```

```
    int rowStartDigit = 1;
```

```
    for (int i = 0; i < n; i++) {
```

```
        int currentDigit = rowStartDigit;
```

```
        for (int j = 0; j < (n - i); j++) {
```

```
            printf("%d", currentDigit);
```

```
            currentDigit = 1 - currentDigit;
```

```
        }
```

```
        printf("\n");
```

```
        rowStartDigit = 1 - rowStartDigit;
```

```
    }
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    if (scanf("%d", &n) != 1) {
```

```
        return 0;
```

```
    }
```

```
    if (n >= 1 && n <= 50) {
```

```
        printBinaryPattern(n);
```

```
    }
```

```
    return 0;
}
```

7.#Binary Pattern 6

```
#include <stdio.h>

void printPattern(int n) {
    int mid = n / 2;

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i == j ||
                i + j == n - 1 ||
                i == mid ||
                j == mid) {

                printf("* ");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
}
```

```
int main() {
    int n;

    if (scanf("%d", &n) != 1) {
        return 0;
    }
}
```

```

    }
    if (n >= 3 && n <= 99 && n % 2 != 0) {
        printPattern(n);
    }

    return 0;
}

```

8.#Asterisk Pattern: X with Horizontal and Vertical Cut

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

void printAlphabetPyramid(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= (n - i); j++) {
            printf(" ");
        }
        char ch = 'A';
        for (int j = 1; j <= (2 * i - 1); j++) {
            printf("%c ", ch);
            ch++;
        }
        printf("\n");
    }
}

```

```

int main() {
    int n;

```

```
    if (scanf("%d", &n) != 1) {  
        return 0;  
    }  
    if (n >= 1 && n <= 12) {  
        printAlphabetPyramid(n);  
    }  
  
    return 0;  
}  
10.#Filled Alphabet Pyramid Pattern
```

#page 4

```
#include <stdio.h>  
#include <string.h>  
#include <math.h>  
#include <stdlib.h>  
int main() {  
    int n;
```

```
if (scanf("%d", &n) != 1) {  
    return 0;  
}
```

```
int arr[n];  
long long sum = 0;  
for (int i = 0; i < n; i++) {  
    if (scanf("%d", &arr[i]) == 1) {  
        sum += arr[i];  
    }  
}  
printf("%lld\n", sum);
```

```
return 0;  
}
```

1.#Sum of Elements in Array 6

```
#include <stdio.h>  
#include <string.h>  
#include <math.h>  
#include <stdlib.h>
```

```
int main() {  
    int n;  
    if (scanf("%d", &n) != 1) {  
        return 0;  
    }
```

```
int arr[n];
```

```

long long evenElementSum = 0;
long long oddElementSum = 0;
long long evenIndexSum = 0;
long long oddIndexSum = 0;

for (int i = 0; i < n; i++) {
    if (scanf("%d", &arr[i]) == 1) {
        if (arr[i] % 2 == 0) {
            evenElementSum += arr[i];
        } else {
            oddElementSum += arr[i];
        }
        if (i % 2 == 0) {
            evenIndexSum += arr[i];
        } else {
            oddIndexSum += arr[i];
        }
    }
}

printf("%lld %lld %lld %lld\n", evenElementSum, oddElementSum, evenIndexSum,
oddIndexSum);

return 0;
}

```

2.#Sum Based on Even/Odd Elements and Indexes

```
#include <stdio.h>
```

```
int main() {
```

```

int n;

scanf("%d", &n);

int arr[n];

for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}

for (int i = n - 1; i >= 0; i--) {
    printf("%d", arr[i]);
    if (i > 0) {
        printf(" ");
    }
}

return 0;
}

```

3.#Reverse the Array

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

void swapNeighbors(int arr[], int n) {
    for (int i = 0; i + 1 < n; i += 2) {
        int temp = arr[i];

```

```

        arr[i] = arr[i + 1];
        arr[i + 1] = temp;
    }
}

```

```

int main() {
    int n;

    if (scanf("%d", &n) != 1) {
        return 0;
    }
    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    swapNeighbors(arr, n);
    for (int i = 0; i < n; i++) {
        printf("%d%s", arr[i], (i == n - 1) ? "" : " ");
    }
    printf("\n");

    return 0;
}

```

6.#Swap Neighbour Elements in an Array

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

```

```

#include <stdio.h>
#include <stdlib.h>

typedef struct {
    int value;
    int index;
} Element;

int compare(const void *a, const void *b) {
    Element *e1 = (Element *)a;
    Element *e2 = (Element *)b;
    if (e1->value < e2->value) return -1;
    if (e1->value > e2->value) return 1;
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    Element *nums = (Element *)malloc(n * sizeof(Element));
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i].value);
        nums[i].index = i;
    }

    int target;
    scanf("%d", &target);
    qsort(nums, n, sizeof(Element), compare);
    int left = 0;
    int right = n - 1;

```

```

while (left < right) {
    long long currentSum = (long long)nums[left].value + nums[right].value;

    if (currentSum == target) {
        printf("%d %d\n", nums[left].index, nums[right].index);
        break;
    } else if (currentSum < target) {
        left++;
    } else {
        right--;
    }
}

free(nums);
return 0;
}

```

7.#Two Sum – Print Indices

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>
#include <stdbool.h>

int main() {
    int n;

    if (scanf("%d", &n) != 1) {
        return 0;
    }
}

```

```
}
```

```
int arr[n];
```

```
bool visited[n];
```

```
for (int i = 0; i < n; i++) {
```

```
    scanf("%d", &arr[i]);
```

```
    visited[i] = false;
```

```
}
```

```
for (int i = 0; i < n; i++) {
```

```
    if (visited[i] == true) {
```

```
        continue;
```

```
}
```

```
int count = 1;
```

```
for (int j = i + 1; j < n; j++) {
```

```
    if (arr[i] == arr[j]) {
```

```
        visited[j] = true;
```

```
        count++;
```

```
    }
```

```
}
```

```
printf("%d : %d\n", arr[i], count);
```

```
}
```

```
return 0;
```

```
}
```

10.#Frequency of All Elements

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>
#include <stdbool.h>

int main() {
    int n;

    if (scanf("%d", &n) != 1) {
        return 0;
    }

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for (int i = 0; i < n; i++) {
        bool isDuplicate = false;
        for (int j = 0; j < i; j++) {
            if (arr[i] == arr[j]) {
                isDuplicate = true;
                break;
            }
        }

        if (!isDuplicate) {
            static bool first = true;
```

```

        if (!first) {
            printf(" ");
        }
        printf("%d", arr[i]);
        first = false;
    }
}

printf("\n");

return 0;
}

```

2.#Remove Duplicates from an Array 5

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

int main() {
    int m, n;

    if (scanf("%d %d", &m, &n) != 2) {
        return 0;
    }

    int matrix[m][n];

    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }
}

```

```

for (int i = 0; i < m; i++) {
    long long rowSum = 0;

    for (int j = 0; j < n; j++) {
        rowSum += matrix[i][j];
    }
    printf("%lld\n", rowSum);
}

return 0;
}

```

4.#Row-wise Sum of a 2D Array

```

#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <stdio.h>

int main() {
    int r;
    const int cols = 12;
    if (scanf("%d", &r) != 1) {
        return 0;
    }
    int data[r][cols];
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < cols; j++) {

```

```

        scanf("%d", &data[i][j]);
    }
}

double maxAverage = -1e18;
for (int j = 0; j < cols; j++) {
    long long columnSum = 0;

    for (int i = 0; i < r; i++) {
        columnSum += data[i][j];
    }
    double currentAverage = (double)columnSum / r;
    if (currentAverage > maxAverage) {
        maxAverage = currentAverage;
    }
}
printf("%g\n", maxAverage);

return 0;
}

```

5.#Maximum Column-wise Average in a 2D Array